

YANG HU

Curriculum Vitae

CONTACT

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School of Mathematics, University of Minnesota Twin Cities
Minneapolis, Minnesota, USA

EDUCATION

University of Minnesota Twin Cities

Spring 2026–present

B.A. in Mathematics, in progress.

UMN GPA: 3.8/4.0 (through Spring 2026).

Southern University of Science and Technology

Shenzhen, China

Undergraduate studies in mathematics; transferred to UMN in Spring 2026.

RESEARCH INTERESTS

My research interests lie broadly in representation theory, number theory, and arithmetic geometry. I am particularly interested in representations of finite and p -adic groups, the Weil representation, and Howe correspondence.

MANUSCRIPT IN PREPARATION

Irreducible Representations of Oscillator Semigroups over Finite Fields and the Branching Problem.

Manuscript in preparation, 2026.

RESEARCH AND DIRECTED READING

Oscillator semigroups and finite-field Howe correspondence

2026–present

Research with Dihua Jiang, University of Minnesota.

Investigating irreducible representations of centralizers in Howe's oscillator semigroups and their restrictions to classical groups. Developing explicit branching calculations for orthogonal–symplectic and general linear dual pairs using finite semigroup representation theory, Harish–Chandra induction, Hecke algebras, and Zelevinsky's parametrization.

Representations of p -adic groups and the Weil representation

2026

Directed reading with Dihua Jiang, University of Minnesota.

Studied representations of GL_2 and SL_2 , Kirillov and Whittaker models, and character theory. Preparing expository notes on the Weil representation, metaplectic extensions, and explicit cocycle calculations.

Winter research project

Research with Baiying Liu (Purdue University), undertaken while enrolled at SUSTech.

SELECTED GRADUATE COURSEWORK

University of Minnesota Twin Cities

Fall 2026 (in progress)

Algebraic Geometry; Algebraic Topology; Lie Groups and Lie Algebras; Real Analysis.

University of Minnesota Twin Cities

Spring 2026 (completed)

Abelian Varieties and Their Moduli (Kai-Wen Lan).

CONFERENCES AND SUMMER SCHOOLS

Arithmetic and Algebraic Geometry Bootcamp

August 17–21, 2026

Participant, Northwestern University, Evanston, Illinois.